

Commentary

Virtual Cell Challenge: Toward a Turing test for the virtual cell

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Virtual cells are an emerging frontier at the intersection of artificial intelligence and biology. A key goal of these cell state models is predicting cellular responses to perturbations. The Virtual Cell Challenge is being established to catalyze progress toward this goal. This recurring and open benchmark competition from the Arc Institute will provide an evaluation framework, purpose-built datasets, and a venue for accelerating model development.

Introduction

Rapid advances in single-cell technologies and high-throughput perturbational measurements have motivated renewed efforts to model cell behavior quantitatively. These “virtual cells” are expected to learn the relationship between cell state and function and are intended to predict the consequences of perturbations—such as a gene knockdown or the application of a drug—across cell types and cell contexts. Given the central role of cellular behavior across diverse fields of basic biology and translational research, the impact of such models could be enormous. Initial efforts in the late 1990s and early 2000s aimed to create comprehensive computational models of the cell from first principles but stalled due to limited data availability, oversimplified assumptions, and computational constraints.^{1–5} Today, with massively improved single-cell measurement technologies, richer perturbational datasets, and advances in machine learning, the field is poised to revisit and realize these early ambitions.⁶

However, the absence of standardized benchmarks and evaluations for assessing model quality has limited progress in predictive cell state modeling. Motivated by the success of the CASP (Critical Assessment of protein Structure Prediction, <https://predictioncenter.org>) competition for protein folding, a structured, recurring

community-wide competition can catalyze progress and align efforts around common goals. It is important to recognize that predictive models of cell behavior will differ from protein structure prediction in many ways. These models must account for additional complexity—such as the cell type, genetic background, and context of a cell—as well as the cellular phenotype being measured and predicted. This community challenge will serve to benchmark progress using carefully designed evaluation datasets and formalize standards with the goal of influencing the evolution of the field.

Open-source competitions can lead to rapid progress

Despite recent growth in the development of predictive models for single-cell perturbation responses, the research community working to model cellular states and behavior currently lacks principled shared evaluation frameworks that reflect the capacity of the model for biologically meaningful generalization across perturbations and cell types. Perturbation model performance is affected by substantial technical sources of variability in many existing datasets, including experimental noise introduced during the coupling of genetic perturbations with single-cell transcriptomic profiling, as well as the limited reproducibility of perturbation effects across inde-

pendent experiments.⁷ Without standardized benchmarks and purpose-built evaluation datasets that evolve in real time alongside developments in the field, it is difficult to evaluate whether models are capturing generalizable biological structure rather than dataset-specific patterns.

Past public competitions have laid important groundwork for benchmarking the prediction of cellular responses to perturbations, but key gaps remain. The Broad Institute’s Cancer Immunotherapy Data Science Grand Challenge, for instance, focused on predicting broad phenotypic shifts in T cells but did not assess responses at the resolution of gene expression.⁸ More recently, the 2023 NeurIPS-Kaggle competition advanced the field by evaluating gene expression changes in response to small-molecule perturbations in immune cells.⁹ Building on these efforts, the field is now ready for a benchmarking competition that assesses gene expression response to genetic perturbations—a task that is central to biological understanding of cell function. Unlike small molecules, genetic perturbations have precisely defined targets, making them ideal for probing causal gene-function relationships, even though the resulting transcriptional changes are often more subtle and harder to predict.



To bridge these gaps, we are launching the Virtual Cell Challenge (<https://virtualcellchallenge.org>), an annual challenge that evaluates AI models of cellular response (Figure 1A). For the inaugural challenge, we have generated a dedicated dataset measuring single-cell responses to perturbations in a human embryonic stem cell line (H1 hESC). This set of perturbations was carefully curated to span a broad range of phenotypic responses, and experimental parameters were optimized to maximize the reproducibility of observed effects. The H1 hESC dataset generated for the Virtual Cell Challenge also contributes to the broader effort to establish experimental and quality control standards for reproducible, high-quality single-cell functional genomics (scFG) data.⁷ Such standards will enable progress and set the community up for building on a solid foundation.¹⁰

This effort is intended to create a level playing field, drive community engagement, and accelerate progress by providing high-quality benchmark datasets, a public leaderboard, and a mechanism for reproducible and fair comparison.

Format of the Virtual Cell Challenge Task

Predictive models can be trained to generalize along several axes. Two core dimensions are (1) generalization across biological context (e.g., cell type, cell line, culture conditions, or even *in vivo* versus *in vitro* settings) and (2) generalization to novel genetic and/or chemical perturbations, including their combinations.

The first Virtual Cell Challenge, launching in 2025, will focus on context generalization as a highly challenging real-world task: participants will predict the effects of single-gene perturbations in a held-out cell type (the H1 hESC; Figure 1B). The transcriptomic consequences of these genetic perturbations have been previously reported in at least one other cellular context.¹¹ This reflects a common experimental reality: testing all perturbations in every context is impractical due to cost, yet accurate context-specific predictions are crucial because responses depend on factors like cell type, state, differentiation stage, culture conditions, and genetic background.

Given that most published single-cell genetic perturbation datasets span only

a handful of cell lines, true zero-shot generalization to new cell states is likely premature. A more appropriate strategy at this stage is few-shot adaptation, where a subset of perturbations in the new cellular context is provided to guide model generalization. To support this, we provide expression profiles for a subset of perturbations measured directly in H1 hESCs, enabling participants to adapt their models before predicting responses to the remaining unseen perturbations in the same cell type.

Evaluations

Evaluation metrics should reflect the core purpose of a virtual cell: simulating cellular behavior via *in silico* experiments—specifically, predicting gene expression responses to genetic perturbations. The main readouts from such an experiment are post-perturbation expression counts and differentially expressed gene sets. Based on these criteria, we have designed three metrics to evaluate model performance (Figure 1C). The first two, differential expression score and perturbation discrimination score, are complementary. The differential expression score evaluates how accurately a model predicts differential gene expression, a key output of most scFG experiments and an essential input for downstream biological interpretation. The perturbation discrimination score measures a model's ability to distinguish between perturbations by ranking predictions according to their similarity to the true perturbational effect, regardless of their effect size.¹²

These two metrics capture distinct and nonoverlapping aspects of performance. For instance, a naive model that consistently predicts the same set of commonly differentially expressed genes from the training data might achieve a reasonable differential expression score. However, its perturbation discrimination score would be random or at the lower bound. Conversely, a model that successfully distinguishes perturbation effects based on subtle variation in an embedding space might perform well on perturbation discrimination. However, this is unlikely to produce biologically meaningful differentially expressed gene sets, which limits its practical utility as a replacement for experimental measurements.

To ensure that predictions are also evaluated across all genes, including

those that are not differentially expressed, we include a third metric: mean absolute error (MAE). While MAE is less biologically interpretable, it captures overall predictive accuracy and provides a global view of model performance across the entire gene expression profile. Together, these three metrics provide a comprehensive evaluation framework. We will use a combined score that appropriately weighs each component, and we will enforce minimum thresholds on all metrics to promote a balanced performance, discouraging models that perform well on one metric at the expense of the others. We will also report model ranking on each individual metric to aid with interpretation of results and highlight models that excel along any one dimension.

Datasets

The first Virtual Cell Challenge will be evaluated on Arc Institute-generated data for H1 hESCs. Additional training data are available from our Virtual Cell Atlas (<https://arcinstitute.org/tools/virtualcellatlas>) and public perturbation datasets.

New Arc-generated benchmark dataset.

For this challenge, we used scFG to generate approximately 300,000 single-cell RNA-sequencing (scRNA-seq) profiles by silencing 300 carefully selected genes using CRISPR interference (CRISPRi)¹³ (Figure 1D). We selected these perturbations to represent a broad range in strength of downstream changes (number of significant differentially expressed genes) and phenotypic diversity of perturbation effects in our initial broad, low-depth screen in H1 hESCs while ensuring adequate representation in existing public datasets. The evaluation dataset was generated using the Flex scRNA-seq platform with high target cell coverage (median of ~1,000 cells per perturbation) and depth of sequencing (over 50,000 unique molecular identifiers [UMIs] per cell). We adopted 10x Genomics' Flex chemistry for this challenge because it offers a more scalable workflow, reduces batch effects through cell fixation, and achieves improved residual knockdown efficiency. Participants will receive a transcriptomic reference of unperturbed H1 hESCs, a training set consisting of single-cell profiles for 150 gene perturbations (~150,000 cells), a validation set of 50 genes whose perturbations are used

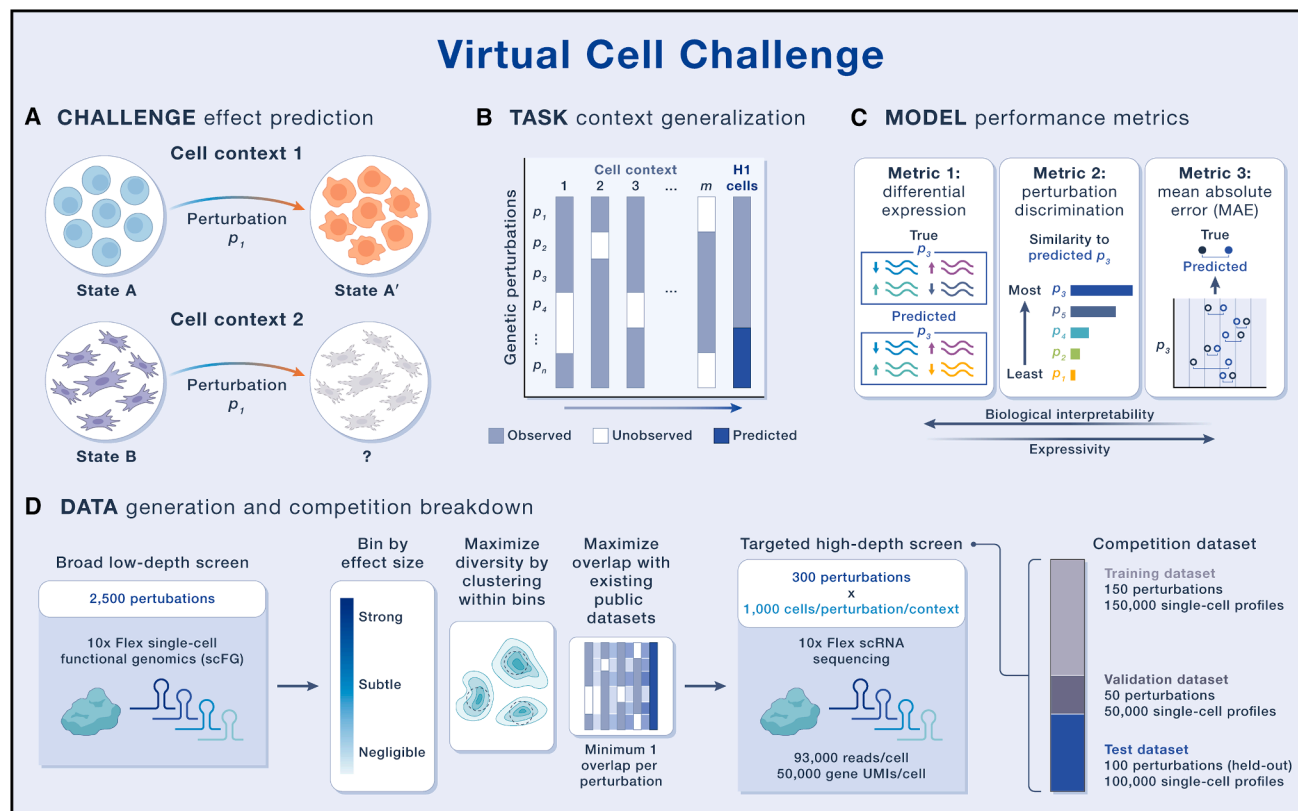


Figure 1. Overview of the Virtual Cell Challenge

(A) In the Virtual Cell Challenge, participants will develop models to predict the transcriptional impact of genetic and chemical perturbations on cell populations. Given an unperturbed cell state, the task involves inferring the gene expression response of cells subjected to specific perturbations.

(B) Cell context is the dimension along which models will be tested for generalization; having observed the impact of genetic perturbations across several cell contexts, and given a set of perturbations in the new cell context (H1 human embryonic stem cell line), can a model predict the impact of other perturbations in H1?

(C) The challenge employs three metrics to evaluate performance: (1) the accuracy of the model in predicting differentially expressed genes in response to a given genetic perturbation, (2) the similarity of the predicted gene expression changes to the experimental ground truth relative to the other perturbations, and (3) the mean absolute error between the model-predicted gene expression and true expression following perturbation for all genes.

(D) A schematic describing our data generation approach. Starting with a single-cell functional genomics (scFG) dataset of 2,500 CRISPR interference (CRISPRi) perturbations in H1 cells, we selected 300 perturbations based on the strength and the diversity of responses. While also maximizing overlap with existing perturbation datasets that can be used for model training, we then generated a high-quality competition dataset centered on these 300 genetic perturbations with greater read depth and a median of 1,000 cells per perturbation. We performed single-cell profiling using 10x Genomics Flex chemistry. Median cells per perturbation, mean reads per cell, and median gene UMIs per cell are reported. The competition dataset is broken down into training data released at the time of competition launch, validation data used for the live leaderboard, and the held-out test dataset used for final scoring of models.

to create a live leaderboard during the competition, and a final test set of 100 held-out perturbations, released a week prior to the submission deadline. Final rankings will be based solely on performance on the held-out test set.

Arc Virtual Cell Atlas. This recently released resource is composed of *scBaseCount*¹⁴ and *Tahoe-100M*,¹⁵ the largest collection of publicly available observational and perturbational scRNA-seq datasets, respectively. With single-cell data from over 350 million cells and counting, this repository offers the community an

ever-expanding training set for the next generation of virtual cell models.

Public perturbation datasets. We will also provide preprocessed versions of selected publicly available perturbation datasets that overlap with the perturbations that are included in the test set.^{11,16–18} While this selection is not exhaustive, these few large datasets are intended to support model training, since observing the effects of target perturbations in alternate cell contexts may be beneficial during model training.

Community engagement

The Virtual Cell Challenge was created to bring together researchers around the unified goal of building better models of cell behavior. Participants will receive regular updates via Arc's Virtual Cell Challenge portal and be featured in leaderboard snapshots throughout the competition. By combining community engagement with rigorous benchmarking, our hope is that the competition will (1) surface best practices and critical data considerations—from training set quality to sequencing depth and platform

choice—that will support more reproducible and robust research efforts; (2) identify methodological bottlenecks, clarify data and/or model choices, and converge on more reproducible standards for perturbation modeling; and (3) highlight the importance of training data quality, cell coverage, sequencing depth, and technology choice. By underscoring these issues, guidelines for more reproducible and principled data generation efforts might emerge.

Future directions

This effort is grounded in ongoing data generation at the Arc Institute, aimed at producing reference-quality datasets to advance the empirical foundations of predictive cell modeling. We hope it also encourages similar initiatives across the community to generate high-quality reference datasets, whether from scRNA-seq or other modalities, that contribute to building more comprehensive and generalizable models of cellular behavior. While the inaugural challenge focuses on genetic perturbation prediction in a single cell type, future challenges will expand to combinatorial perturbations and cross-cell-type generalization. Looking ahead, a comprehensive modeling of cellular behavior will require integrating diverse biological modalities—including transcriptional, proteomic, and epigenetic—across temporal and spatial dimensions in multicellular systems. Similarly, the inverse of perturbation response prediction, which aims to identify optimal perturbations that achieve a desired effect, is an important capability for Virtual Cell models with clear therapeutic relevance for future challenges. We view this initiative as an evolving test bed for advancing quantitative modeling, where both datasets and evaluation frameworks are designed to adapt as the field deepens its understanding of biological complexity. Metrics will be refined based on insights from prior years' results. We invite the broader scientific community to shape and strengthen future versions of the Virtual Cell Challenge.

Conclusion

Virtual cells are poised to become foundational tools for biology, and to ensure that they reach their potential, we need

clear and rigorous evaluations. The Virtual Cell Challenge aims to provide just that: a fair, open challenge to surface the best models, clarify the state of the art, and engage the community. We invite the community to engage with this first iteration and help better define the contours of predictive cellular modeling as a scientific discipline.

DECLARATION OF INTERESTS

D.P.B. is a Google Advisor.

H.G. is a cofounder of Exai Bio, Tahoe Therapeutics, and Therna Therapeutics; serves on the board of directors at Exai Bio; and is a scientific advisory board member for Verge Genomics and Deep Forest Biosciences.

P.D.H. is a cofounder of Monet AI, Terrain Biosciences, and Stylus Medicine; serves on the board of directors at Stylus Medicine; is a board observer at EvolutionaryScale and Terrain Biosciences; is a scientific advisory board member at Arbor Biosciences and VedaBio; and is an advisor to NFDG, Varda Space, and Vial Health.

L.A.G. is a cofounder of nChroma Bio and a scientific advisory board member of Myllia Biotechnology.

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